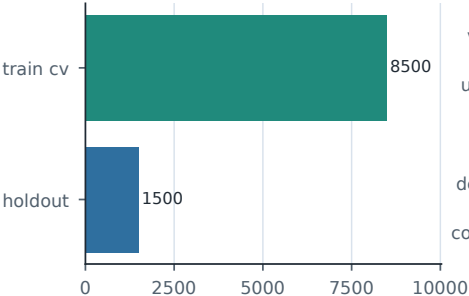


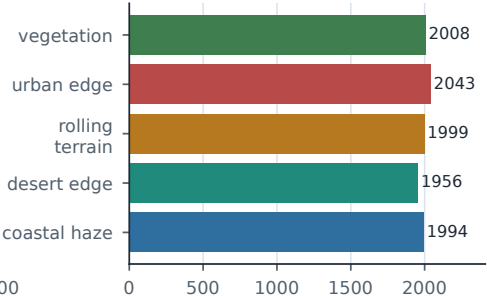
Scenario balance and leakage diagnostics

Scenario groups are locked before phase expansion; the evidence keeps label, site, range, aspect, noise, and hard-negative family counts together.

Split



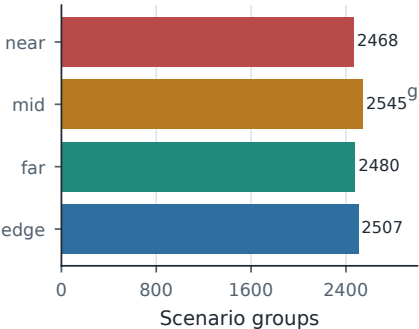
Site



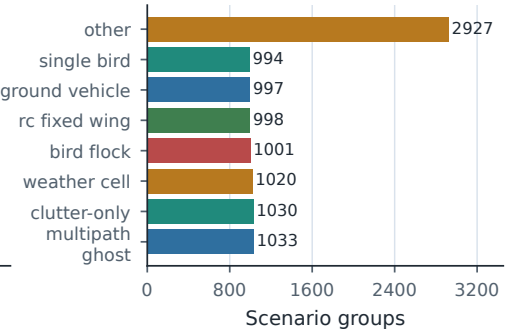
Leakage checks

canary test: detected and blocked
stratum baseline AP: 0.005
metadata baseline AP: 0.004
label shuffle AP: 0.006

Range



Hard-negative



Marginal imbalance

aspect: 0.000
hard_negative_role: 0.032
noise: 0.000
range: 0.000
site: 0.000
split: 0.390